

4. **ESC and Battery Considerations**:

- Since you're already using **30A ESCs**, **5-inch props** should not overtax the ESCs, especially if they're calibrated and running correctly.
- If you decide to continue using larger props, ensure that your **ESCs are rated for higher current** (i.e., **40A or higher**), but ideally, using a better prop size (matching the motor's RPM capabilities) will be more effective than just relying on larger ESCs.

Summary of Next Steps:

1. **Switch to smaller props**: Try **5x4.5** or **5x5** props (for 3S) or **6x4** props (for 4S).
2. **Test without props**: Before flying with props, test the motors to ensure they are running at the expected RPM without load.
3. **Ensure balanced ESCs**: Make sure all ESCs are the same rating (e.g., **30A** or **35A** for consistency) to avoid power imbalances.
4. **Recalibrate ESCs**: Make sure to recalibrate your ESCs after changing props or ESCs.

5. ****Check thrust-to-weight****: With smaller props, your drone should have a better ****thrust-to-weight ratio****, which is key for stable flight and takeoff.

Let me know how it goes after making these adjustments or if you need help selecting the right props or ESCs!

General ESC Sizing Rule of Thumb:

Motor KV	Battery Voltage (S)	Max Current (A)	ESC Rating (A)
1000-1500	3S (11.1V)	15-25A	20-30A
1500-2500	3S (11.1V)	25-40A	30-40A
2200-2800	3S (11.1V)	30-45A	35-50A
2500-3500	4S (14.8V)	35-50A	40-60A

ESC Current Rating Formula:

$$\text{ESC Rating} \geq \text{Maximum Motor Current Draw} \times 1.25 \text{ to } 1.3$$

Estimated Motor Current Draw I_{motor} (in Amps):

$$I_{motor} = \frac{P_{motor}}{V_{battery}}$$

Where:

- P_{motor} is the power consumed by the motor (in watts)
- $V_{battery}$ is the battery voltage (e.g., 3S = 11.1V, 4S = 14.8V, etc.)

You can estimate the power consumption P_{motor} from:

$$P_{motor} = KV \times V_{battery} \times I_{motor}$$

Formula for RPM:

$$RPM = KV \times \text{Battery Voltage}$$

Where:

- KV is the motor's KV rating (in this case, 2200KV).
- Battery Voltage is the nominal voltage of your battery (for a 3S battery, it's 11.1V; for a 4S battery, it's 14.8V, and so on).

$$\begin{aligned} RPM &= KV \times V \\ &= 2200 \times 11.1 \\ RPM &= 24,420 \text{ RPM} \end{aligned}$$

for determining the plane's stability and how it will behave in flight.

The CG should generally be located around **25% to 33% of the wing's chord** from the leading edge for most stable flight. If the CG is too far forward, the plane will be nose-heavy, and if it's too far aft, it will be tail-heavy and less stable. Based on the dimensions you provided, we would expect that the CG is reasonably close to the optimal position for stability.

Summary of Calculations:

1. **Wing Loading:** 39.9 N/m^2 — Moderate, indicating a relatively stable, lightweight plane.
2. **Thrust-to-Weight Ratio:** 1.44 — More than enough thrust for good performance and climbing ability.
3. **Power Consumption:** Estimated at **49 W** for level flight at 20 m/s.
4. **Flight Time:** Estimated at **13.8 minutes** with a 1000mAh LiPo battery.
5. **CG:** Falls within a reasonable range (4.6–5.2 cm from the leading edge), indicating good stability.

This should give you a solid understanding of your RC plane's performance characteristics. If you need further details or adjustments, feel free to ask!